

Little Paws



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We feel so gratified for preparing ourselves for future professional obligations in such a reputable institute.

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1: Document Version

Table 1 - Version of Document

Version	Date	Description	Authors
V1.1	26 January 2022	LittlePaws	Zurriat Naveed (18F-0221)
V1.2	21 April 2022		M. Hamza Mahmood (18F-0421)
V1.3	29 June 2022		M. Abubakar (18F-0151)

1.1: Introduction

LittlePaws is a web-based application where users can buy or sell their cats. It will contain all the regular features that an e-commerce website has, like Add to cart, ratings, transactions, etc. The unique feature of this website is that the buyer can perform value-based search as well as image-based search. It will be done using image processing techniques.

For this purpose, the buyer will have to upload the image of the cat, to find the desired cat breed. The system will process that image and the results will be shown to the user. In Pakistan there are many persons that did not know the proper name of cat, but they want to buy that cat to overcome this problem this website will help them. It will reduce the human effort and it will save the time of the buyer.

1.1.1: Motivation

The motivation behind this project is to create easiness for the cat buyers. And help a newly obsessed cat lover, who doesn't have any prior knowledge about the cats; to find his/her desired cat.

Through this technology, the trend of buying cats will also increase. It will be an easy source of income for too many people.

1.1.2: Objectives

The objectives of Littlepaws are:

- To reduce human effort
- To save the time of the buyer

1.1.3: Scope

- It is a web-based application which includes buyer and seller panel. The buyer panel will be able to purchase cats. Similarly, seller panel will be able to sell their cats on the website.
- Module for users (A user can perform both roles i.e., buyer and seller. The user will be either buyer or a seller. If the user wants to purchase a cat than it will be a buyer. On the other hand, if the user wants to sell than it will be a seller.)

1.1.4: Constrains

- User should have knowledge of using web application.
- User must have stable internet connection.
- User must be registered.
- User should be aware of knowledge of buying and selling on website.

1.2: Positioning

Littlepaws fills the market needs and wants. To mitigate against the financial burden, littlepaws restructured its business model to operate as a 'marketplace' where sellers could set up virtual stores at littlepaws by listing their products and setting prices. This way our application grows, and we can earn money through this application in the market.

1.2.1: Business Opportunity

Once this website is implemented our team can proudly say that this application would help a lot of people to buy and sell their cats.

Secondly, this system would also help them to pay online.

So, overall, our website will be eligible of doing both things. It will be a selling system as well as buying system based on image processing. Where buyers can search in both ways i.e., by value and by image.

1.2.2: Problem Statement

Sometimes, the buyer doesn't have the proper knowledge about the cat. That's why it becomes almost a difficult task.

The time of the buyer is wasting during the search as, buyer doesn't have relevant information about the cat.

1.2.3: Solution

This project Littlepaws is providing the solution to this problem of buying the cat by uploading its image or by writing its name.

So, to solve the above-mentioned problem, buyer must upload the image of the cat. And the result will be displayed. No prior knowledge is required.

1.3: Stakeholder and User Descriptions

1.3.1: Stakeholder Summary

Table 2 - Stakeholder Summary

Name	Description
Admin	The admin panel.
Buyer	The buyer panel (i.e. to buy cats)
Seller	The Seller panel (i.e. to sell cats)

1.3.2: User Summary

Table 3 - User Summary

Stakeholder	Goal	Problem	Concerns
Admin	To handle all the process.	To handle the process.	Generate notifications.
Buyer	to buy cats	the buyer doesn't have the proper knowledge about the cat	By uploading image.
Seller	to sell cats	The seller wants to sell their cats.	Add product with complete details.

2. Software requirement specification

2.1 List of features

2.1.1 Buyer panel

- **Login**

There will be a login form through which the buyer will get logged in to the website.

- **Register**

To buy the cat the buyer must be registered on the website. There will be a registration form through which the buyer will register itself.

- **Search by value**

The buyer should use the search by value filter to buy its desired cat.

- **Search by image**

The buyer should search the desired cat by uploading the image of the cat.

- **Add to cart**

The buyer should add all the desired cats into the cart before preceding to the payment. In cart the details of the desired cats will be shown to the buyer.

- **Make payment**

The buyer will be able to pay online through an external payment gateway.

- **Raise conflict**

The buyer should raise conflict in case of any problem or mishap.

- **Rating**

The buyer should rate on all its orders after the order completion.

2.1.2 Admin panel

- **Resolve conflict**

The admin should be able to resolve all the conflicts occurred on the website.

- **Show conflict**

All the conflicts occurred on the website should be visible to the admin.

2.1.3 Seller panel

- **Login**

There will be a login form through which the seller will get logged in to the website.

- **Register**

To buy the cat the seller must be registered on the website. There will be a registration form through which the seller will register itself.

- **Raise conflict**

The seller should raise conflict in case of any problem or mishap.

- **Show orders**

All the seller's orders should be visible to the seller.

- **Add catalog**

The seller should be able to add and post multiple catalogs on the website.

- **Update catalog**

The seller should be able to update all its catalogs on the website.

- **Delete catalog**

The seller should be able to delete all its catalogs from the website.

2.2 Functional Requirements

2.2.1 *Buying/selling Cats*

- The customer shall be able to select his/her desired cat specie and then shall be able to buy the selected cat species.
- And similarly, the sellers will be selling their cats on the website.

2.2.2 *Search by Value*

- The customer should search the desired cat specie by passing the name of the cat.
- The customer should search the desired cat specie by passing some other characteristic like color, eye color and size of the cat.

2.2.3 *Search by Image*

- If the customer does not have any prior information about the cat. Then, customer may search the desired cat by uploading the image of the cat.

2.2.4 *Add To Cart*

- The customer should add all the selected cats in the shopping cart.

2.2.5 *Make payment*

- After selecting the desired cat, the customer can make the payment for the selected cats.

2.2.6 *Rating*

- The customer should give the rating after the purchasing the cat in the form of ratings the stars.
- And the seller should response back to the feedback of the customers.

2.2.7 Raise/Resolve Conflict

- In case of any problem, the customer and seller should be able to raise the conflict.
- And the admin should be resolving the conflicts faced by the sellers and buyers, using its conflict resolving responsibility.

2.2.8 Manage Catalog

- The seller should be able to manage the catalog i.e., add cat, remove cat, and update cat.
- The customer should search the catalog. And then buy from that catalog.

2.3 Quality Attributes

2.3.1 Usability

The application should be easy to use and shall be appropriate for the target market of public.

2.3.2 Maintainability

The application shall be designed for ease of maintenance. All the data stored, should be aligned and modifiable with proper authorized person.

2.4 Non-functional requirements

2.4.1 Availability

- The website shall be available 24/7.
- And in case of any downfall, the recovery time of the website will be no more than 30 minutes.

2.4.2 Learnability

- The frontend and functionality of the website will be easy enough that any laymen can easily learn how to use the website. And can perform different operations on the website.

2.4.3 User-friendly

- The user and the company will have their email and id numbers therefore after verification from the system they can continue working.
- E-wallet can only be access by the company for updating purpose.

2.4.4 Response time

- The response time of any task or activity on the website will always be less than 30 seconds.

2.5 Other requirements

2.5.1 Safety requirements

- This website shall include uploading the image and will get the output i.e. the matched image; therefore, our application will be in safe state.

2.5.2 Security requirements

- The customer will have his email and id numbers therefore after verification from the system they can continue working.

3 Use case

3.1 Use case system Diagram:

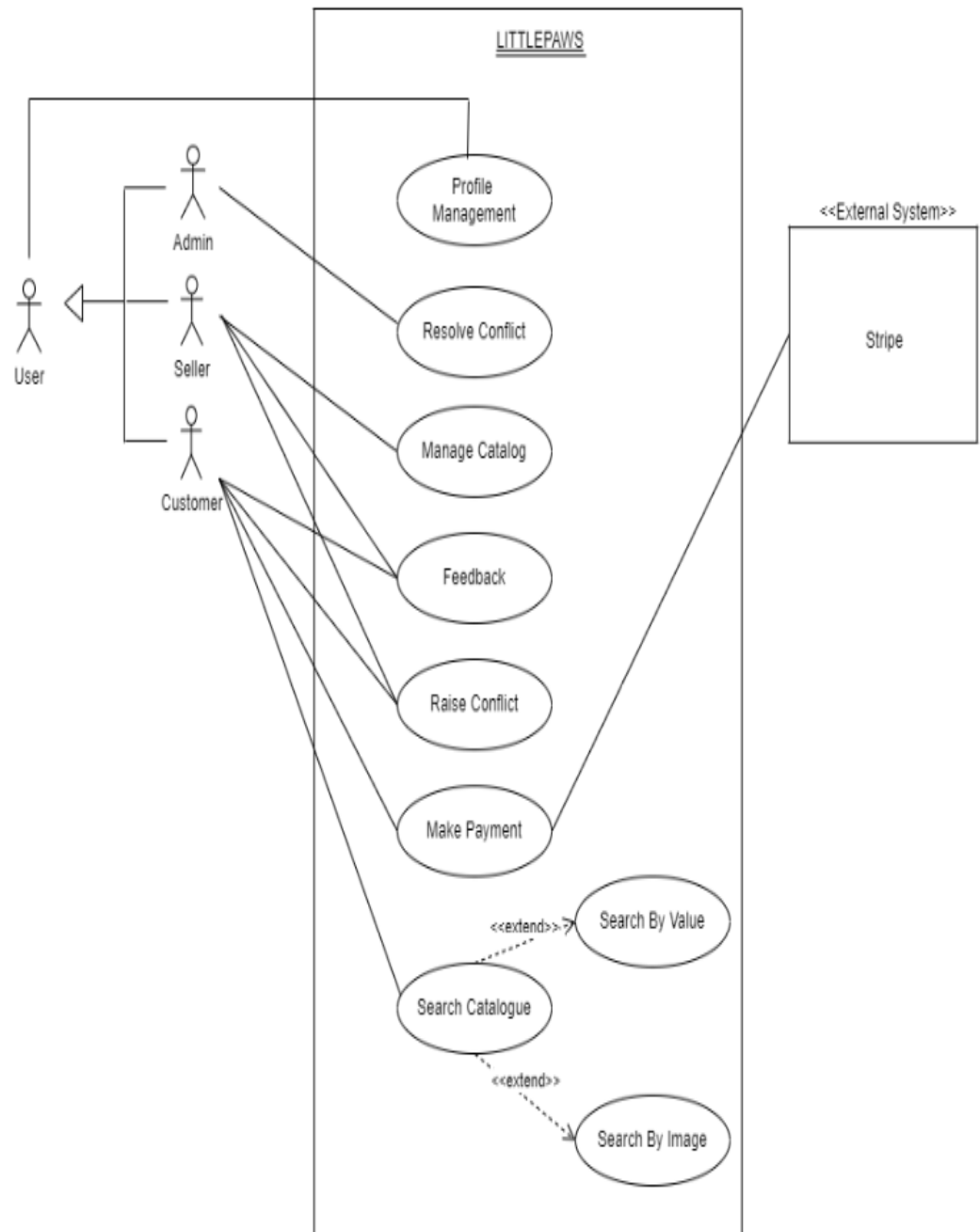


Figure 1: Use case Diagram

3.2 High Level Use Cases

3.2.1 *Manage Account*

Use Case Name	Manage Account
Actors	Admin, Seller, Customer
Type	Primary
Description	Users will be able to manage their accounts.

3.2.2 *Resolve Conflict*

Use Case Name	Resolve Conflict
Actor	Admin
Type	Primary
Description	The admin will receive all the conflicts and is responsible to resolve those conflicts.

3.2.3 *Feedback*

Use Case Name	Feedback
Actors	Seller, Customer
Type	Primary
Description	The customer will be able to give the feedback after purchasing of the cat. And the seller can view as well as response back to the feedback.

3.2.4 Raise Conflict

Use Case Name	Raise Conflict
Actors	Seller, Customer
Type	Primary
Description	The customer and seller can raise conflict if they are facing any problem.

3.2.5 Make Payment

Use Case Name	Make Payment
Actors	External System, Customer
Type	Secondary, Primary
Description	The customer will pay the payment of the order. The payment will be done through an external payment system i.e., Jazz Cash API.

3.2.6 Manage Catalog

Use Case Name	Manage Catalog
Actors	Seller
Type	Primary
Description	The seller can manage its catalog that is add catalog, update catalog and delete catalog.

3.2.7 Search Catalog

Use Case Name	Search Catalog
Actors	Customer
Type	Primary
Description	The customer can search the catalogs. Search can be performed by passing some values or by uploading the image of the cat.

4 Iteration 1

4.1 Expanded Use Cases

4.1.1 *Raise Conflict*

Use Case Name	Raise Conflict	
Actor	Buyer, Seller	
Type	Primary	
Description	The seller and customer can raise the conflict if they are facing any issue.	
Pre-conditions	The seller and customer face any issue.	
Post-conditions	The conflict will be resolved by the admin. Customer and seller will be informed through email.	
Normal Flow	1. The seller or customer face an issue. 2. The seller/customer will raise its conflict by clicking on the “Conflict” button and describe the problem. 3. A conflict number will be issued to the customer/seller. 4. When the conflict is resolved, the customer/seller will be informed through an email.	
Alternative Scenario/s	2.1. correct login and registered.	
Frequency of Use	Low	

4.1.2 Make Payment

Use Case Name	Make Payment
Actor	Customer, External System
Type	Primary, Secondary
Description	The customer will pay the payment. And the payment will be handled through an external payment gateway.
Pre-conditions	The customer must have cats in the cart.
Post-conditions	The customer will receive the order details and the payment slip.
Normal Flow	1. The order details and amount against the order will be displayed to the customer. 2. The customer will pay the amount by clicking the “pay now” button. 3. After that the customer will be redirected to the payment window.
Alternative Scenario/s	3.1 The transection may fail due to exceptions.
Frequency of Use	Normal

4.1.3 Manage Catalog

Use Case Name	Manage Catalog
Actor	Seller
Type	Primary
Description	The seller will be able to manage all his/her catalogs.
Pre-conditions	The seller must be logged in and have details of the cats.
Post-conditions	The catalog will be posted on the website.
Normal Flow	1. The seller will login its account. 2. The seller can add catalog by entering the details of the cat. And post it by clicking on the “Add Catalog” button. 3. The seller may delete and update the already created catalogs.
Alternative Scenario/s	1.1. Incorrect login and credential.
Frequency of Use	Normal

4.1.4 Search Catalog

Use Case Name	Search Catalog
Actor	Customer
Type	Primary
Description	The customer can search any cat by uploading the picture of the cat or by entering the values of the cat.
Pre-conditions	The user must have a clear picture of cat or know the values of cat.
Post-conditions	The user will find the cat if it is available on website otherwise “not found” will be displayed.
Normal Flow	1. The customer will upload image of cat or input the values of cat. 2. The results will be displayed to the customer, whether it was searched by image or by value.
Alternative Scenario/s	2.1 No results will be shown in case of unavailability of that cat.
Frequency of Use	Normal

4.1.5 Resolve Conflict

Use Case Name	Resolve Conflict
Actor	Admin
Type	Primary
Description	The admin will resolve all the conflicts that are raised from the seller/customer.
Pre-conditions	Admin should be login on the website.
Post-conditions	The admin sends resolved conflict email to the seller/customer.
Normal Flow	1. The admin gets all the conflicts from users. 2. Select a specific conflict. 3. Resolve the issue of the user. 4. Send confirmation email to the user.
Alternative Scenario/s	1.1. Admin don't find any conflict in conflict box.
Frequency of Use	Normal

4.1.6 Manage Account

Use Case Name	Manage Account	
Actor	Admin , Seller , Customer	
Type	Primary	
Description	User can manage their accounts in terms of reset password, change profile details.	
Pre-conditions	The user must have an account on the website.	
Post-conditions	The profile changes will be saved on the website.	
Normal Flow	1. 2. 3. 4. 5.	The user will login the account. Click on “Profile” button. Edit profile details. Save changes of the profile. The users can change the password of their account.
Alternative Scenario/s	1.1.	The user may forget its account credentials.
Frequency of Use	Normal	

4.1.7 Feedback

Use Case Name	Feedback	
Actor	Seller, Customer	
Type	Primary	
Description	Customer can give feedback on the product and seller give response back to the feedback.	
Pre-conditions	The customer must have purchased any cat from the website.	
Post-conditions	The feedback will be posted on the catalog.	
Normal Flow	1. The customer login his/her account on the website. 2. Search the cat from the complete order list. 3. Enter the feedback and post it.	
Alternative Scenario/s	None	
Frequency of Use	Normal	

4.2 Activity diagram

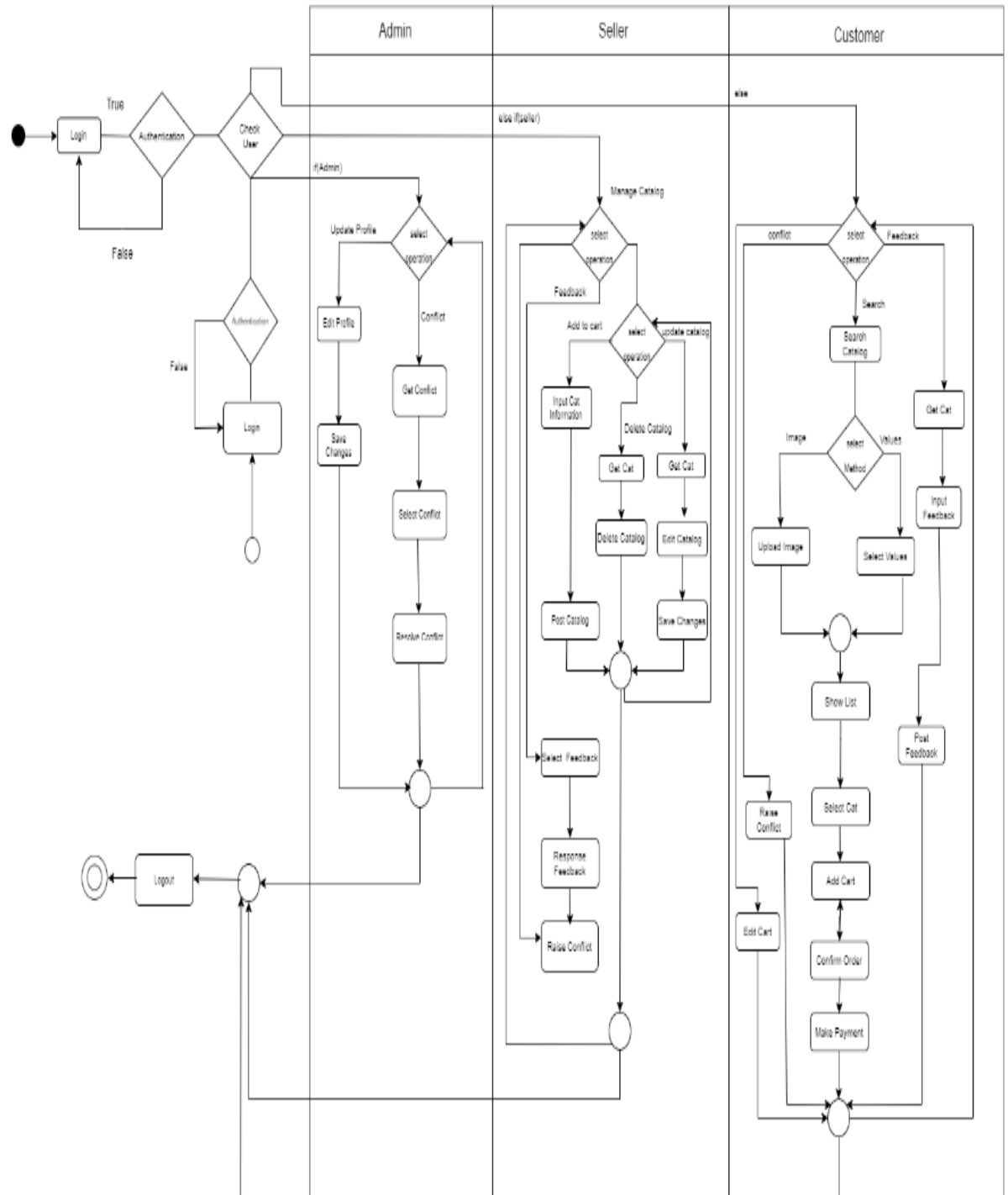


Figure 2 : Activity Diagram

4.3 Domain model

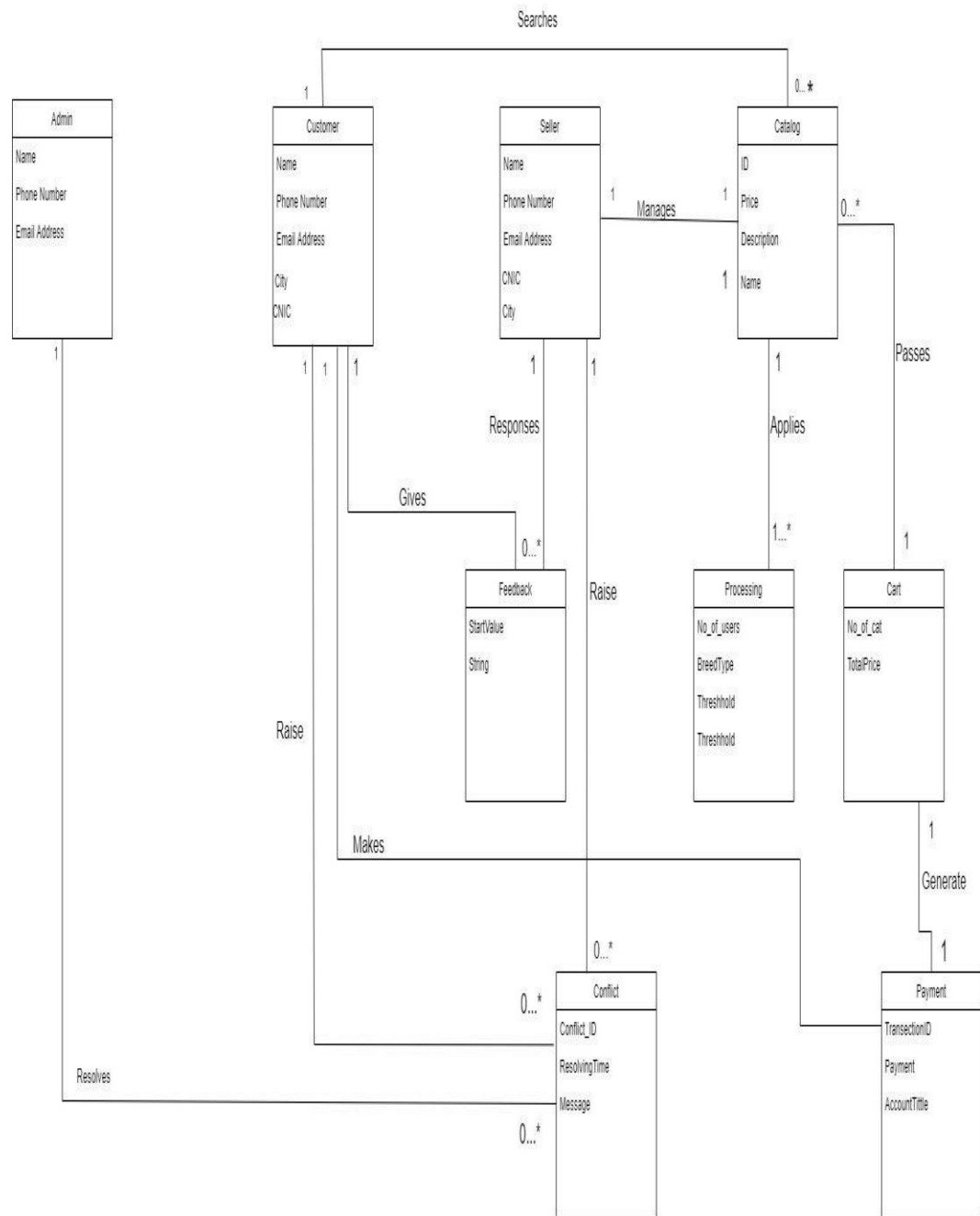


Figure 3: Domain model

4.4 System sequence diagram

4.4.1 Manage Catalogue

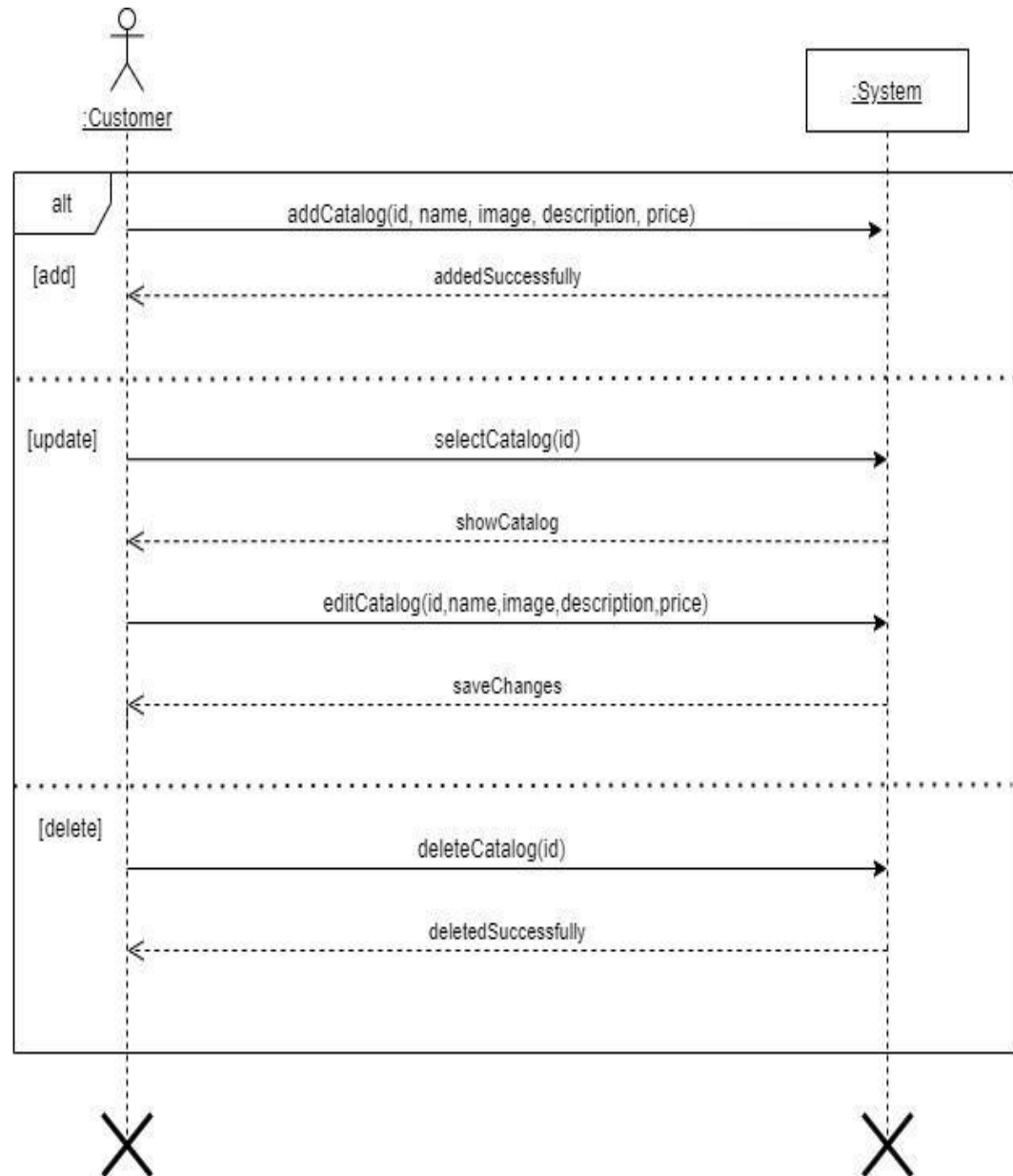


Figure 4 SSD1

4.4.2 Make Payment

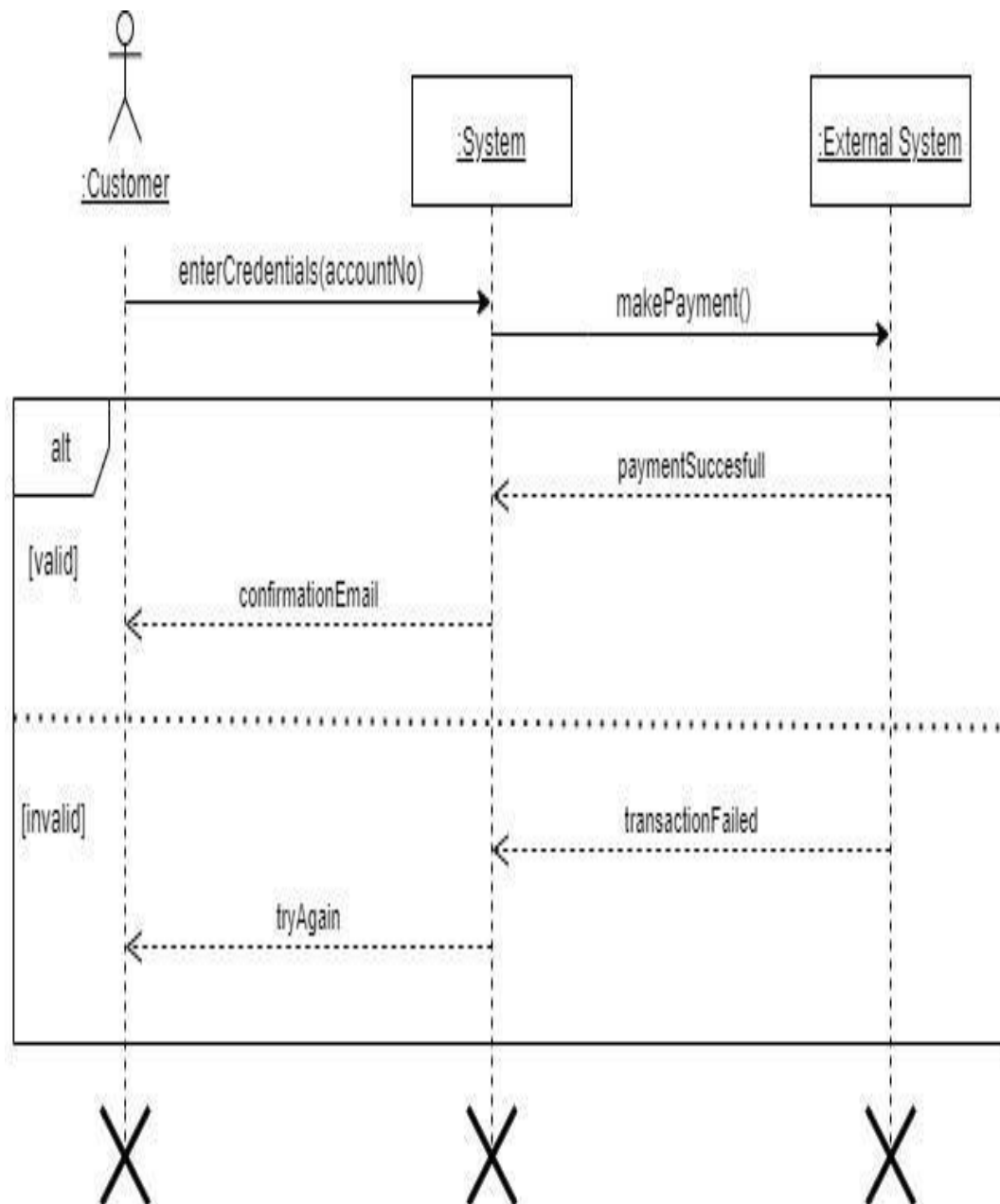


Figure 5 SSD 2

4.4.3 Search Catalog

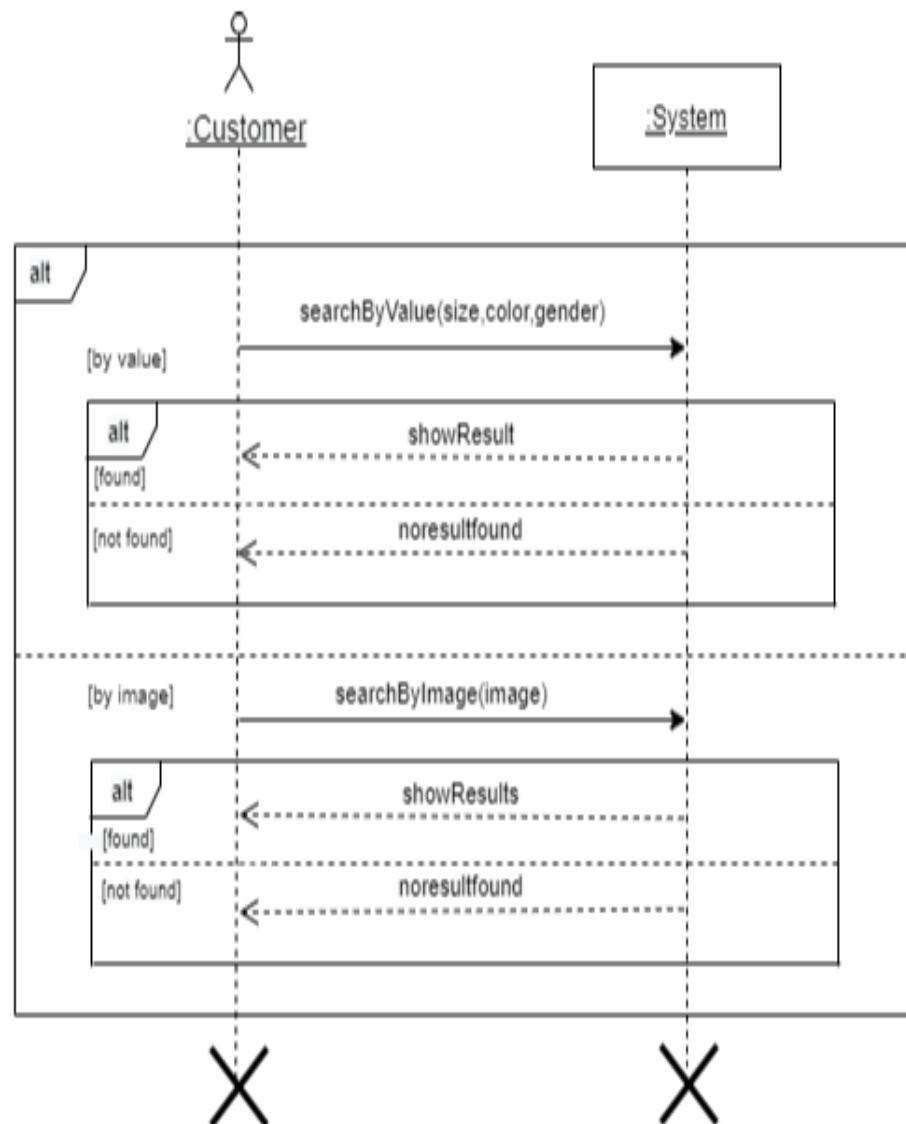


Figure 6 SSD3

4.4.4 Manage Account

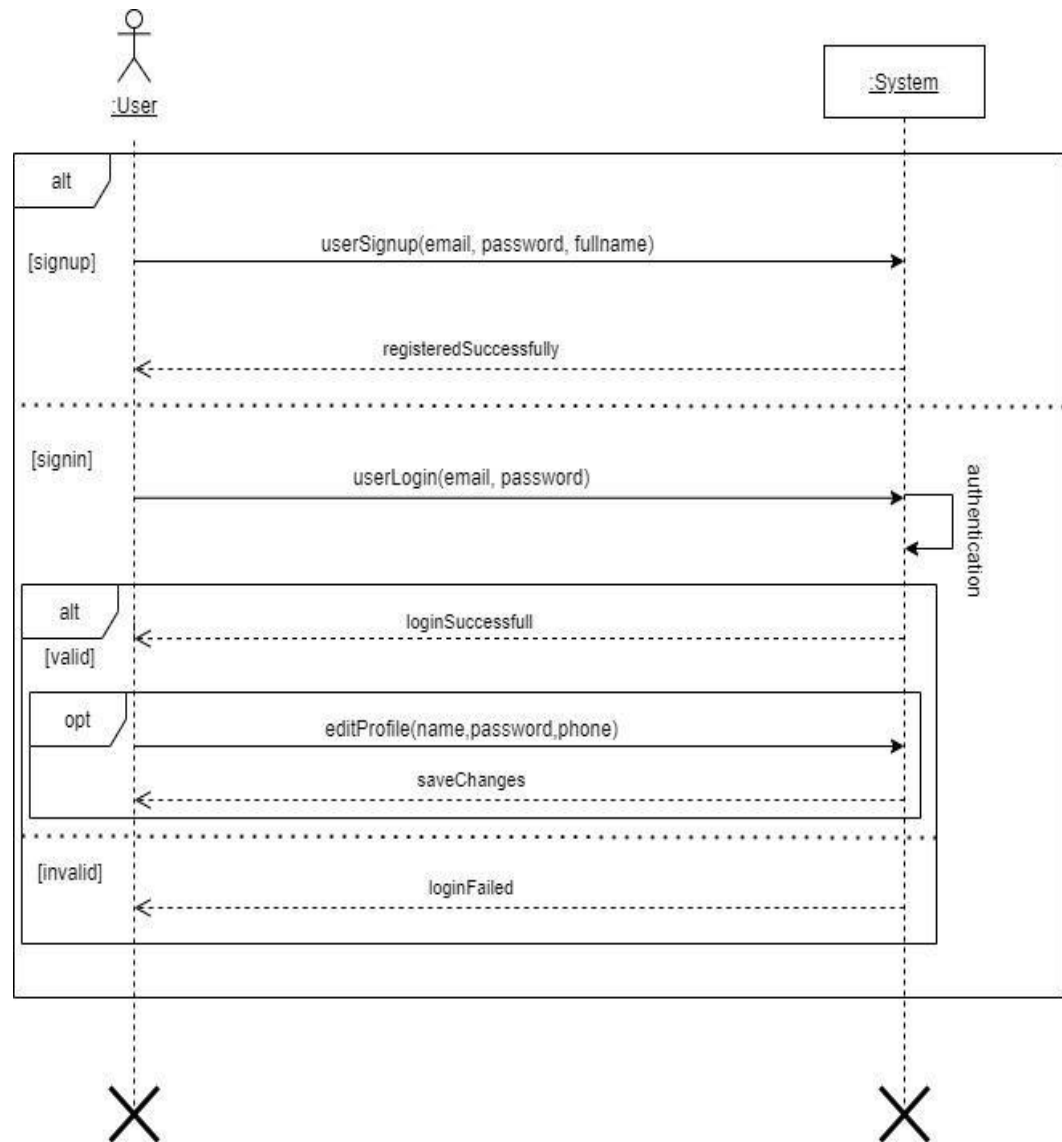


Figure 7 SSD 4

4.4.5 Manage Cart

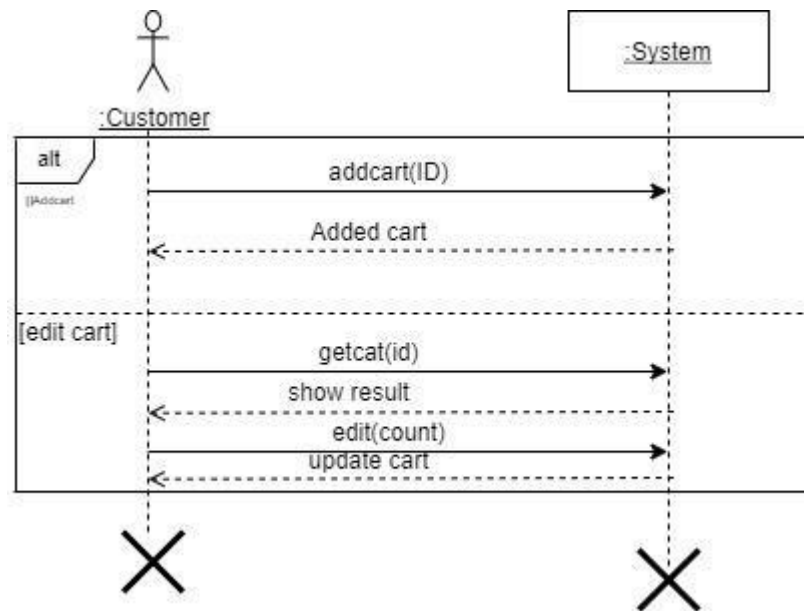


Figure 8 SSD 5

4.4.6 Raise Conflict

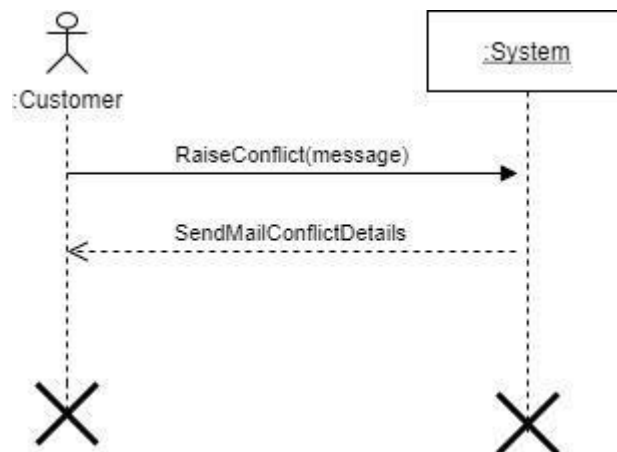


Figure 9 SSD 6

4.4.7 Resolve Conflict

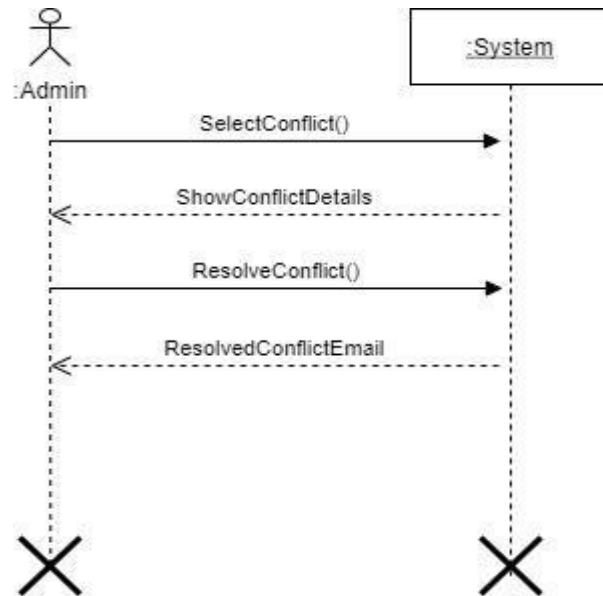


Figure 10 SSD 7

4.4.8 Give Feedback

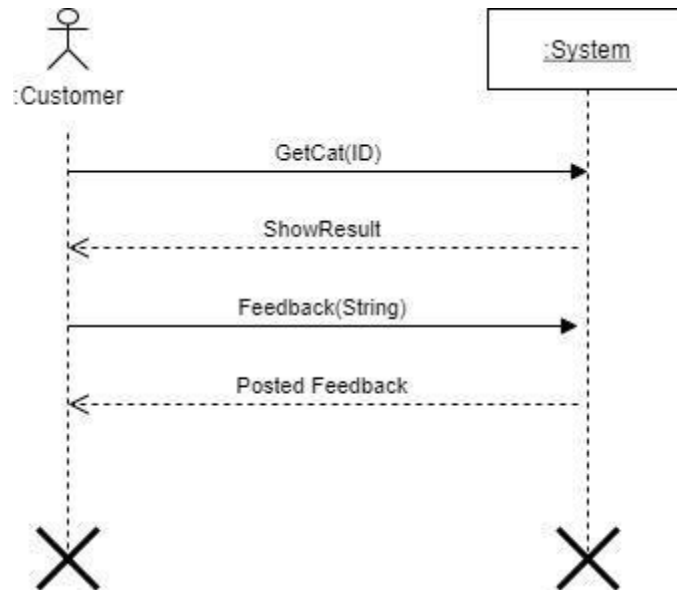


Figure 11 SSD 8

4.5 Operation Contracts

4.5.1 Contract OC1: add Catalog

Operation	addCatalog()
Cross References	Use case: Manage Catalog
Preconditions	Seller must be logged in.
Post-conditions	Catalog class instance is created. Catalog class attributes id, name, description, image, price modified. Catalog class is associated with seller class.

4.5.2 Contract OC2: add Cart

Operation	addCart()
Cross References	Use case: Search Catalog
Preconditions	Customer must be logged in.
Post-conditions	Cart class instance is created. Cart class attributes Cart.noOfCats modified. Cart class is associated with Catalog class.
Operations:	Search Product (Products Name, Categorization,)
References:	Use Case: Product
Preconditions:	Login process is underway
Postconditions:	An instance of Product class is created Product.name to search product Product was assigned product ID Product type for recycle

4.5.3 Contract OC3: Feedback

Operation	feedback ()
Cross References	Use case: Feedback
Preconditions	User must be logged in.
Post-conditions	Customer class instance is created. Customer.comment attribute modified. Customer class is associated with User class.

4.5.4 Contract OC4: Enter Credentials

Operation	enterCredentials()
Cross References	Use case: Make Payment
Preconditions	Customer must be logged in.
Post-conditions	Payment class instance is created. Payment.account, Payment.title, Payment.amount modified. Payment class is associated with Customer class.

4.5.5 Contract OC5: resolve Conflict

Operation	resolveConflict()
Cross References	Use case: Resolve Conflict
Preconditions	Admin must be logged in.
Post-conditions	Admin class instance is created. Admin.status attribute modified. Admin class is associated with conflict class.

4.5.6 Contract OC6: raise Conflict

Operation	raiseConflict ()
Cross References	Use case: Raise Conflict
Preconditions	Seller/Customer must be logged in.
Post-conditions	Conflict class instance is created. Conflict.conflictId and Conflict.conflictDescription attribute modified. Conflict class is associated with customer or seller class.

4.5.7 Contract OC7: search By Image

Operation	searchByImage()
Cross References	Use case: Search Catalog
Preconditions	Customer must be logged in.
Post-conditions	Processing class instance is created. Processing.threshold, Processing.features attributes modified. Processing class is associated with Catalog class.

4.6 Sequence diagrams

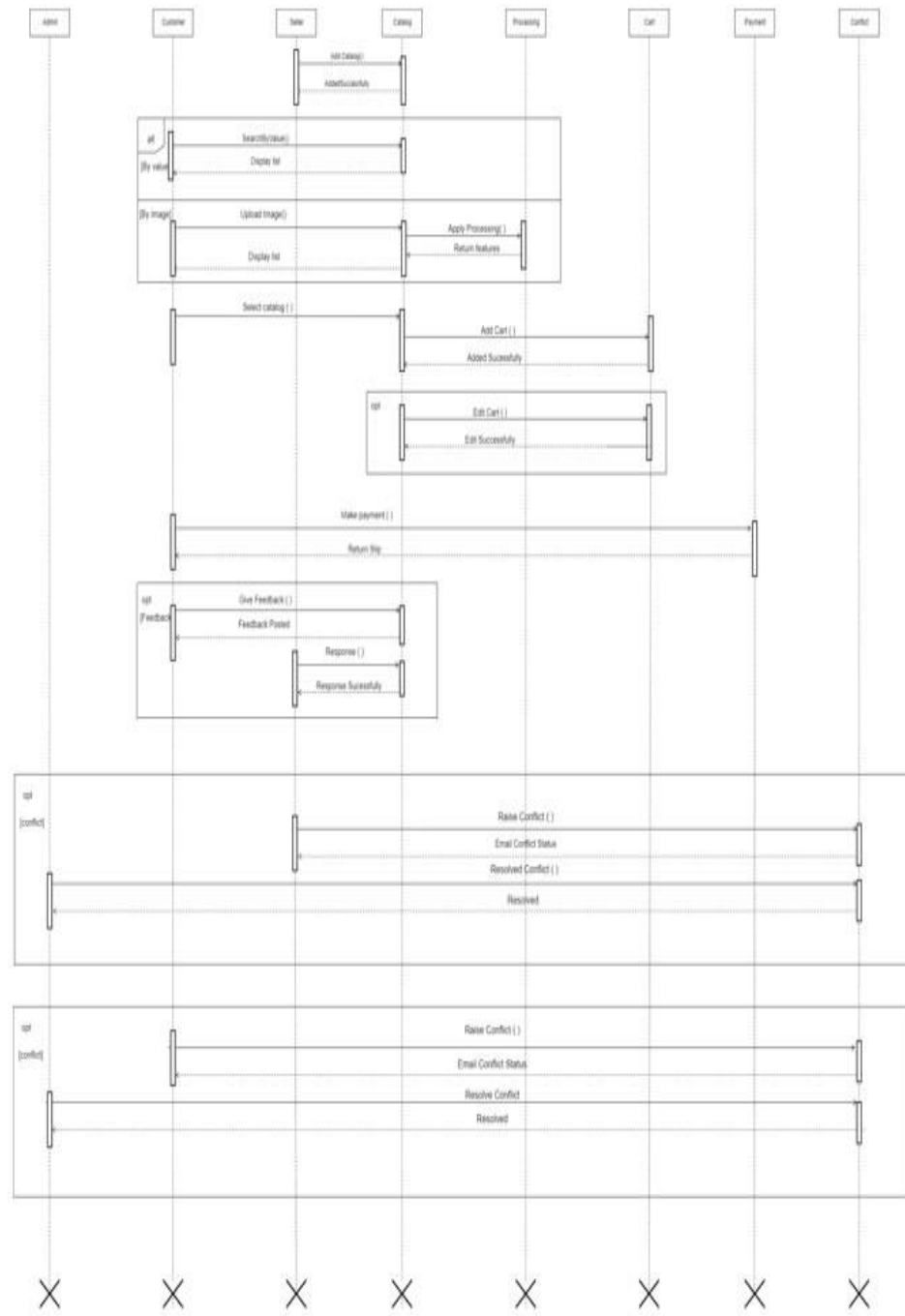


Figure 12 System Sequence Diagram

4.7 Class diagram

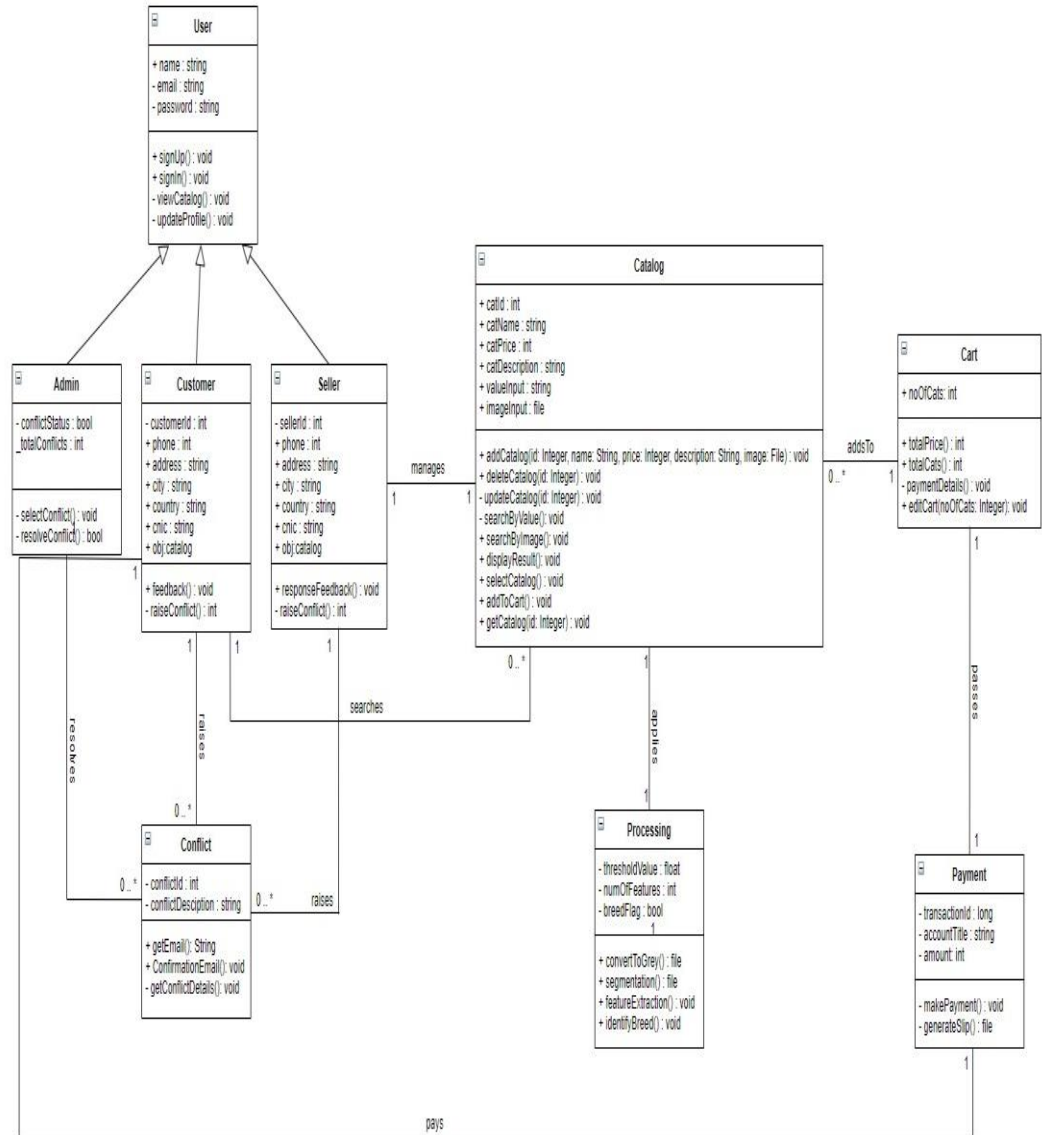


Figure 13 Class Diagram

4.8 Architecture diagram

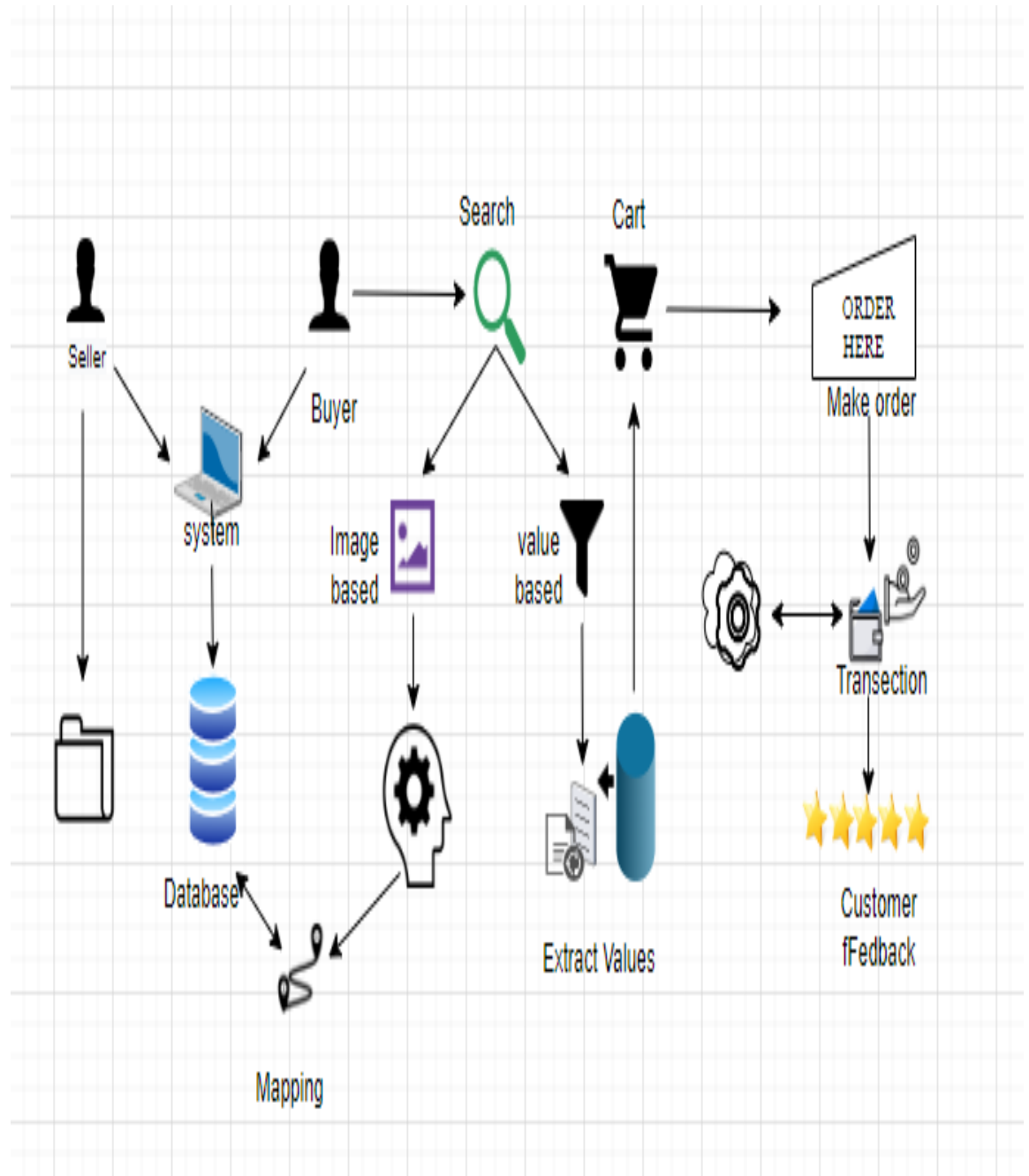


Figure 14 Architecture Diagram

5. References

- [1] <https://www.kaggle.com/ma7555/cat-breeds-dataset>
- [2] <https://www.petfinder.com/>
- [3] <https://www.agentpet.com/>
- [4] <https://catvills.com/cat-breed-identifier-app-reviews/>
- [5] <https://apps.apple.com/us/app/id1450069587>
- [6] <https://www.investopedia.com/terms/e/escrow.asp>